TD/TP 2 – Fundamental Mathematical Concepts

Step-by-step solutions

Toolbox used throughout

Let $\{e_t\}$ be white noise: $\mathbb{E}[e_t] = 0$, $\mathrm{Var}(e_t) = \sigma_e^2$, and $\mathrm{Cov}(e_t, e_s) = 0$ for $t \neq s$.

(W) Weak stationarity. A process $\{Y_t\}$ is (weakly) stationary if

(i) $\mu_t = \mathbb{E}[Y_t]$ is constant in t;
(ii) $\gamma_{t,t-k} = \mathrm{Cov}(Y_t, Y_{t-k})$ depends only on the lag k, not on t.

(B) Bilinearity of covariance.

$$\mathrm{Cov}\Big(\sum_i a_i X_i,\ \sum_j b_j Z_j\Big) = \sum_i \sum_j a_i b_j \,\mathrm{Cov}(X_i, Z_j).$$

Constants do not change covariances: $\mathrm{Cov}(X + c, Z) = \mathrm{Cov}(X, Z)$.

(R) Recipe for any MA-type process. Expand Y_t and Y_{t-k} as sums of e's; by (B) and white noise, only pairs with the *same time index* survive, each contributing (coef)×(coef)×σ_e^2.

Notation. $\rho_k = \gamma_k/\gamma_0$, and $\nabla Y_t = Y_t - Y_{t-1}$.

Exercise 2.4

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$Y_t = e_t + \theta e_{t-1}$ with $\theta = 3$ or $\theta = 1/3$. (a) Find the ACF in both cases. (b) Can we tell which θ is correct from estimates of ρ_k?

(a)

Step 1 – Mean. $\mathbb{E}[Y_t] = \mathbb{E}[e_t] + \theta\mathbb{E}[e_{t-1}] = 0$, constant.

Step 2 – Variance ($k = 0$). e_t and e_{t-1} are uncorrelated, so

$$\gamma_0 = \mathrm{Var}(e_t) + \theta^2 \,\mathrm{Var}(e_{t-1}) = \sigma_e^2(1 + \theta^2).$$

Step 3 – Lag 1.

$$\gamma_1 = \mathrm{Cov}(e_t + \theta e_{t-1},\ e_{t-1} + \theta e_{t-2}).$$

Expanding, the only matching index is e_{t-1} (coefficient θ in the first, 1 in the second): $\gamma_1 = \theta\sigma_e^2$.

Step 4 – Lag $k \geq 2$. Y_t involves $\{e_t, e_{t-1}\}$ and Y_{t-k} involves $\{e_{t-k}, e_{t-k-1}\}$. For $k \geq 2$ these sets do not overlap, so $\gamma_k = 0$.

Step 5 – ACF.

$$\rho_0 = 1, \qquad \rho_1 = \frac{\theta}{1 + \theta^2}, \qquad \rho_k = 0 \ (k \geq 2).$$

Step 6 – Plug in.

$$\theta = 3: \ \rho_1 = \frac{3}{1+9} = \frac{3}{10}; \qquad \theta = \tfrac{1}{3}: \ \rho_1 = \frac{1/3}{1 + 1/9} = \frac{1/3}{10/9} = \frac{3}{10}.$$

Answer

In both cases $\rho_1 = 0.3$ and $\rho_k = 0$ for $k \geq 2$. The process is stationary (constant mean, covariances depend only on k) and the two ACFs are *identical*.

Key idea

In general $\dfrac{\theta}{1+\theta^2} = \dfrac{1/\theta}{1+1/\theta^2}$ (multiply top and bottom by $1/\theta^2$). So θ and $1/\theta$ always give the same ACF.

(b)

Step 1 – What does the data tell us? With mean 0 and $\mathrm{Var}(Y_t) = 1$ known, everything we can estimate about second-order structure is ρ_k, via the sample ACF r_k.

Step 2 – Compare the two models. Under $\theta = 3$ we need $\sigma_e^2(1+9) = 1 \Rightarrow \sigma_e^2 = 0.1$; under $\theta = 1/3$ we need $\sigma_e^2(1+1/9) = 1 \Rightarrow \sigma_e^2 = 0.9$. Both models then have *exactly* the same mean (0), the same variance (1) and the same autocorrelations ($\rho_1 = 0.3$, rest 0). If the noise is Gaussian, the joint distribution of $(Y_1, \ldots, Y_n)$ is fully determined by mean and covariances, so the two models produce *the same distribution* for the data.

Step 3 – Conclusion. r_k estimates ρ_k, which is the same under both hypotheses. However large n is, $r_1 \to 0.3$ in both cases.

Answer

No, we cannot determine θ from $\hat{\rho}_k$: the two parameter values are *not identifiable* because they give identical ACFs (and, for Gaussian noise, identical distributions). In practice one picks $\theta = 1/3$, the *invertible* representation ($|\theta| < 1$), so that e_t can be written as a convergent sum of present and past Y's.

Exercise 2.5

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$Y_t = 5 + 2t + X_t$, $\{X_t\}$ zero-mean stationary with autocovariance γ_k. Find (a) the mean function, (b) the autocovariance function, (c) is $\{Y_t\}$ stationary?

(a) By linearity of expectation and $\mathbb{E}[X_t] = 0$:

$$\mu_t = \mathbb{E}[Y_t] = 5 + 2t + \mathbb{E}[X_t] = 5 + 2t.$$

(b) $5 + 2t$ is deterministic, and constants do not change covariance (rule B):

$$\mathrm{Cov}(Y_t, Y_s) = \mathrm{Cov}(5 + 2t + X_t,\ 5 + 2s + X_s) = \mathrm{Cov}(X_t, X_s) = \gamma_{t-s}.$$

(c) The covariance condition holds (it depends only on $t - s$), but the mean $5 + 2t$ changes with t.

Answer

$\mu_t = 5 + 2t$, $\mathrm{Cov}(Y_t, Y_s) = \gamma_{t-s}$. $\{Y_t\}$ is **not stationary**, because its mean function depends on time (a linear trend). It is "stationary around a trend" (trend-stationary).

Exercise 2.8

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$\{Y_t\}$ stationary with autocovariance γ_k. Show that $W_t = c_1 Y_t + c_2 Y_{t-1} + \cdots + c_n Y_{t-n+1}$ is stationary.

Write compactly $W_t = \sum_{i=1}^{n} c_i Y_{t-i+1}$ and let $\mu = \mathbb{E}[Y_t]$.

Step 1 – Mean.

$$\mathbb{E}[W_t] = \sum_{i=1}^{n} c_i \mathbb{E}[Y_{t-i+1}] = \mu \sum_{i=1}^{n} c_i,$$

which does not depend on t.

Step 2 – Covariance at lag k. Using bilinearity (B):

$$\mathrm{Cov}(W_t, W_{t-k}) = \mathrm{Cov}\left(\sum_{i=1}^{n} c_i Y_{t-i+1},\ \sum_{j=1}^{n} c_j Y_{t-k-j+1} \right) = \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \,\mathrm{Cov}(Y_{t-i+1}, Y_{t-k-j+1}).$$

Step 3 – Use stationarity of Y. The time difference is $(t-i+1) - (t-k-j+1) = k+j-i$, so

$$\mathrm{Cov}(Y_{t-i+1}, Y_{t-k-j+1}) = \gamma_{k+j-i} \quad \Longrightarrow \quad \mathrm{Cov}(W_t, W_{t-k}) = \sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j \, \gamma_{k+j-i}.$$

Answer

The mean is constant and the covariance depends only on k (the variable t has disappeared). Hence $\{W_t\}$ is stationary. A *finite linear filter of a stationary process is stationary.*

Exercise 2.9

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$Y_t = \beta_0 + \beta_1 t + X_t$, X_t zero-mean stationary. (a) Show Y_t is not stationary but $W_t = \nabla Y_t$ is. (b) If $Y_t = \mu_t + X_t$ with μ_t a polynomial of degree d, show $\nabla^m Y_t$ is stationary for $m \geq d$ and nonstationary for $0 \leq m < d$.

(a)

Step 1 – Y_t is not stationary. $\mathbb{E}[Y_t] = \beta_0 + \beta_1 t$, which varies with t whenever $\beta_1 \neq 0$ (the case of interest).

Step 2 – Compute the difference.

$$W_t = Y_t - Y_{t-1} = (\beta_0 + \beta_1 t + X_t) - (\beta_0 + \beta_1 (t-1) + X_{t-1}) = \beta_1 + X_t - X_{t-1}.$$

The trend has collapsed to the constant β_1.

Step 3 – Mean. $\mathbb{E}[W_t] = \beta_1$, constant.

Step 4 – Covariance.

$$\begin{aligned} \mathrm{Cov}(W_t, W_{t-k}) &= \mathrm{Cov}(X_t - X_{t-1},\ X_{t-k} - X_{t-k-1}) \\ &= \mathrm{Cov}(X_t, X_{t-k}) - \mathrm{Cov}(X_t, X_{t-k-1}) - \mathrm{Cov}(X_{t-1}, X_{t-k}) + \mathrm{Cov}(X_{t-1}, X_{t-k-1}) \\ &= \gamma_k - \gamma_{k+1} - \gamma_{k-1} + \gamma_k = 2\gamma_k - \gamma_{k+1} - \gamma_{k-1}. \end{aligned}$$

(Alternatively: this is 2.8 with $c_1 = 1, c_2 = -1$, plus a constant.)

Answer

$W_t = \nabla Y_t$ has constant mean β_1 and autocovariance $2\gamma_k - \gamma_{k+1} - \gamma_{k-1}$ depending only on k, so it is stationary, while Y_t is not.

(b)

Since ∇ is linear, $\nabla^m Y_t = \nabla^m \mu_t + \nabla^m X_t$. We treat the two parts separately.

Step 1 – $\nabla^m X_t$ is always stationary. By 2.8 (with $n = 2$, $c_1 = 1$, $c_2 = -1$), if X_t is stationary then so is ∇X_t, with mean 0. By induction, $\nabla^m X_t$ is stationary with mean 0 for every $m \geq 0$.

Step 2 – ∇ lowers the degree of a polynomial by one. Let $\mu_t = a_d t^d + a_{d-1} t^{d-1} + \cdots + a_0$ with $a_d \neq 0$. By the binomial theorem,

$$t^d - (t-1)^d = t^d - \left(t^d - d\, t^{d-1} + \tbinom{d}{2} t^{d-2} - \cdots\right) = d\, t^{d-1} + (\text{lower powers}).$$

Lower terms t^j $(j < d)$ give at most degree $j - 1 < d - 1$. Hence $\nabla \mu_t$ is a polynomial of degree $d - 1$ with leading coefficient $d\, a_d \neq 0$; a constant $(d = 0)$ is sent to 0.

Step 3 – Iterate. Applying Step 2 m times:

- $m < d$: $\nabla^m \mu_t$ has degree $d - m \geq 1$, leading coefficient $d(d-1)\cdots(d-m+1)a_d \neq 0$;
- $m = d$: $\nabla^d \mu_t = d!\, a_d$, a constant;
- $m > d$: $\nabla^m \mu_t = 0$.

Step 4 – Conclude. $\mathbb{E}[\nabla^m Y_t] = \nabla^m \mu_t$ (since $\mathbb{E}[\nabla^m X_t] = 0$), and the covariances of $\nabla^m Y_t$ equal those of $\nabla^m X_t$ (deterministic parts do not affect covariance), which depend only on the lag.

Answer

For $m \geq d$, the mean $\nabla^m \mu_t$ is constant ($d! a_d$ or 0), so $\nabla^m Y_t$ is stationary. For $0 \leq m < d$, the mean is a non-constant polynomial in t, so $\nabla^m Y_t$ is nonstationary. In words: *differencing d times removes a polynomial trend of degree d.*

Exercise 2.12

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$Y_t = e_t - e_{t-12}$. Show $\{Y_t\}$ is stationary and that, for $k > 0$, $\rho_k \neq 0$ only at $k = 12$.

Step 1 – Mean. $\mathbb{E}[Y_t] = 0 - 0 = 0$.

Step 2 – Variance. $\gamma_0 = \mathrm{Var}(e_t) + \mathrm{Var}(e_{t-12}) = 2\sigma_e^2$.

Step 3 – Lag $k > 0$.

$$\mathrm{Cov}(Y_t, Y_{t-k}) = \mathrm{Cov}(e_t - e_{t-12},\ e_{t-k} - e_{t-k-12}).$$

The four cross terms involve pairs of times $(t, t-k)$, $(t, t-k-12)$, $(t-12, t-k)$, $(t-12, t-k-12)$. With $k > 0$, a pair has equal indices only when $t - 12 = t - k$, i.e. $k = 12$. That term is $\mathrm{Cov}(-e_{t-12}, e_{t-12}) = -\sigma_e^2$. Thus

$$\gamma_k = \begin{cases} 2\sigma_e^2 & k = 0 \\ -\sigma_e^2 & k = 12 \\ 0 & k > 0,\ k \neq 12. \end{cases}$$

Step 4 – ACF. $\rho_{12} = -\sigma_e^2/(2\sigma_e^2) = -\frac{1}{2}$, and $\rho_k = 0$ for other $k > 0$.

Answer

Mean is 0, covariances depend only on $k \Rightarrow$ stationary. For $k > 0$ the only nonzero autocorrelation is $\rho_{12} = -1/2$. (This is a seasonal MA(1) with period 12, typical for monthly data.)

Exercise 2.14

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Find mean and covariance function, and decide stationarity: (a) $Y_t = \theta_0 + te_t$; (b) $W_t = \nabla Y_t$; (c) $Y_t = e_t e_{t-1}$ (e_t normal white noise).

(a) $Y_t = \theta_0 + te_t$

Mean. $\mathbb{E}[Y_t] = \theta_0 + t\mathbb{E}[e_t] = \theta_0$ (constant).

Covariance. $\mathrm{Cov}(Y_t, Y_s) = \mathrm{Cov}(te_t, se_s) = ts\,\mathrm{Cov}(e_t, e_s)$, so

$$\mathrm{Cov}(Y_t, Y_s) = \begin{cases} t^2\sigma_e^2 & t = s \\ 0 & t \neq s. \end{cases}$$

Answer

Mean is constant, but $\mathrm{Var}(Y_t) = t^2\sigma_e^2$ grows with t (the covariance at lag 0 depends on t), so the process is **not stationary**.

(b) $W_t = \nabla Y_t$

Step 1 – Write it out. $W_t = (\theta_0 + te_t) - (\theta_0 + (t-1)e_{t-1}) = te_t - (t-1)e_{t-1}$.

Step 2 – Mean. $\mathbb{E}[W_t] = 0$.

Step 3 – Variance. $\mathrm{Var}(W_t) = t^2\sigma_e^2 + (t-1)^2\sigma_e^2 = \big(t^2 + (t-1)^2\big)\sigma_e^2$.

Step 4 – Lag 1. $\mathrm{Cov}(te_t - (t-1)e_{t-1},\ (t-1)e_{t-1} - (t-2)e_{t-2})$: only e_{t-1} matches, giving $-(t-1)\cdot(t-1)\sigma_e^2$:

$$\mathrm{Cov}(W_t, W_{t-1}) = -(t-1)^2\sigma_e^2.$$

Step 5 – Lag $k \geq 2$. No common e's, so $\mathrm{Cov}(W_t, W_{t-k}) = 0$.

Answer

$\mathbb{E}[W_t] = 0$, $\mathrm{Var}(W_t) = (t^2 + (t-1)^2)\sigma_e^2$, $\mathrm{Cov}(W_t, W_{t-1}) = -(t-1)^2\sigma_e^2$, 0 beyond lag 1. Variance and lag-1 covariance depend on t, so **not stationary**. Note: differencing removes trends in the *mean*, but it cannot fix a variance that grows with time.

(c) $Y_t = e_t e_{t-1}$

Here we need expectations of products, using independence (normal white noise is i.i.d.).

Step 1 – Mean. $\mathbb{E}[Y_t] = \mathbb{E}[e_t]\mathbb{E}[e_{t-1}] = 0$.

Step 2 – Variance. $\mathrm{Var}(Y_t) = \mathbb{E}[e_t^2 e_{t-1}^2] = \mathbb{E}[e_t^2]\mathbb{E}[e_{t-1}^2] = \sigma_e^4$.

Step 3 – Lag 1.

$$\mathrm{Cov}(Y_t, Y_{t-1}) = \mathbb{E}[e_t e_{t-1} \cdot e_{t-1} e_{t-2}] = \mathbb{E}[e_t]\,\mathbb{E}[e_{t-1}^2]\,\mathbb{E}[e_{t-2}] = 0 \cdot \sigma_e^2 \cdot 0 = 0.$$

Step 4 – Lag $k \geq 2$. $\mathbb{E}[e_t e_{t-1} e_{t-k} e_{t-k-1}]$: the index t appears only once, so the expectation factors with $\mathbb{E}[e_t] = 0$, giving 0.

Answer

$\mathbb{E}[Y_t] = 0$, $\gamma_0 = \sigma_e^4$, $\gamma_k = 0$ for $k \neq 0$. The process is **stationary** – in fact it is itself white noise (uncorrelated), although its terms are *not independent*: Y_t and Y_{t-1} share e_{t-1} (e.g. Y_t^2 and Y_{t-1}^2 are correlated). Uncorrelated $\neq$ independent.

Exercise 2.17

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$\{Y_t\}$ stationary, $\bar{Y} = \frac{1}{n}\sum_{t=1}^n Y_t$. Show $\mathrm{Var}(\bar{Y}) = \dfrac{\gamma_0}{n} + \dfrac{2}{n}\displaystyle\sum_{k=1}^{n-1}\Big(1 - \frac{k}{n}\Big)\gamma_k = \frac{1}{n}\sum_{k=-n+1}^{n-1}\Big(1 - \frac{|k|}{n}\Big)\gamma_k$.

Step 1 – Expand the variance as a double sum.

$$\mathrm{Var}(\bar{Y}) = \frac{1}{n^2}\,\mathrm{Cov}\Big(\sum_{t=1}^n Y_t, \sum_{s=1}^n Y_s\Big) = \frac{1}{n^2}\sum_{t=1}^n\sum_{s=1}^n \gamma_{t-s}.$$

Step 2 – Group terms by lag $k = t - s$. Picture the $n \times n$ matrix with entries γ_{t-s}: along each diagonal $t - s = k$ the entry is constant. For $k \in \{-(n-1), \ldots, n-1\}$, the number of pairs (t, s) with $1 \leq t, s \leq n$ and $t - s = k$ is $n - |k|$ (e.g. $k = 0$: n pairs; $k = n - 1$: only $(n, 1)$). So

$$\mathrm{Var}(\bar{Y}) = \frac{1}{n^2}\sum_{k=-n+1}^{n-1}(n - |k|)\gamma_k = \frac{1}{n}\sum_{k=-n+1}^{n-1}\Big(1 - \frac{|k|}{n}\Big)\gamma_k.$$

This is the second form.

Step 3 – Fold negative lags. Pull out $k = 0$ (giving γ_0/n) and use $\gamma_{-k} = \gamma_k$, so lags k and $-k$ give equal terms:

$$\mathrm{Var}(\bar{Y}) = \frac{\gamma_0}{n} + \frac{2}{n}\sum_{k=1}^{n-1}\Big(1 - \frac{k}{n}\Big)\gamma_k.$$

This is the first form.

Answer

Both identities hold. Interpretation: if $\gamma_k = 0$ for $k \neq 0$ (white noise) we recover the classical γ_0/n. Positive autocorrelation *increases* $\mathrm{Var}(\bar{Y})$, so the sample mean is less precise than i.i.d. theory suggests; negative autocorrelation does the opposite.

Exercise 2.23

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$\{Z_t\}$ and $\{Y_t\}$ are independent if, for any times $t_1, \ldots, t_m$ and $s_1, \ldots, s_n$, the vector $(Z_{t_1}, \ldots, Z_{t_m})$ is independent of $(Y_{s_1}, \ldots, Y_{s_n})$. Show that if they are independent stationary processes, $W_t = Z_t + Y_t$ is stationary.

(The definition in the sheet has a missing phrase; the line above is the intended statement.)

Step 1 – Mean. $\mathbb{E}[W_t] = \mathbb{E}[Z_t] + \mathbb{E}[Y_t] = \mu_Z + \mu_Y$, constant.

Step 2 – Covariance.

$$\mathrm{Cov}(W_t, W_s) = \mathrm{Cov}(Z_t, Z_s) + \mathrm{Cov}(Z_t, Y_s) + \mathrm{Cov}(Y_t, Z_s) + \mathrm{Cov}(Y_t, Y_s).$$

Step 3 – Cross terms vanish. By independence of the two processes, Z_t is independent of Y_s, so $\mathrm{Cov}(Z_t, Y_s) = \mathrm{Cov}(Y_t, Z_s) = 0$.

Step 4 – Use stationarity of each.

$$\mathrm{Cov}(W_t, W_s) = \gamma^Z_{t-s} + \gamma^Y_{t-s},$$

which depends only on $t - s$.

Strict-stationarity version (optional). If both are strictly stationary: by independence, the joint law of $(Z_{t_1-k}, \ldots, Z_{t_m-k}, Y_{t_1-k}, \ldots, Y_{t_m-k})$ is the product of the Z-law and the Y-law; each factor is unchanged by the shift k, so the joint law is shift-invariant. $(W_{t_1}, \ldots, W_{t_m})$ is a fixed function of this vector, so its law is shift-invariant too.

Answer

$\mathbb{E}[W_t] = \mu_Z + \mu_Y$ and $\gamma^W_k = \gamma^Z_k + \gamma^Y_k$, so $\{W_t\}$ is stationary. (Weakly, only *uncorrelatedness* of the two processes is actually needed.)

Exercise 2.26

Exercise 2.26

Semivariogram $\Gamma_{t,s} = \frac{1}{2}\mathbb{E}[(Y_t - Y_s)^2]$. (a) Show that for a stationary process $\Gamma_{t,s} = \gamma_0 - \gamma_{|t-s|}$. (b) Show the random walk is intrinsically stationary ($\Gamma_{t,s}$ depends only on $|t - s|$).

(a)

Step 1 – Center. With $\mu = \mathbb{E}[Y_t]$ constant, $Y_t - Y_s = (Y_t - \mu) - (Y_s - \mu)$ (μ's cancel).

Step 2 – Expand the square.

$$\mathbb{E}[(Y_t - Y_s)^2] = \mathbb{E}[(Y_t - \mu)^2] + \mathbb{E}[(Y_s - \mu)^2] - 2\mathbb{E}[(Y_t - \mu)(Y_s - \mu)] = \gamma_0 + \gamma_0 - 2\gamma_{t-s}.$$

Step 3 – Halve, using $\gamma_{t-s} = \gamma_{|t-s|}$.

Answer

$\Gamma_{t,s} = \gamma_0 - \gamma_{|t-s|}$.

(b)

Step 1 – Random walk. $Y_t = e_1 + e_2 + \cdots + e_t$. Recall it is *not* stationary: $\mathrm{Var}(Y_t) = t\sigma_e^2$.

Step 2 – Difference. Take $t > s$. All terms up to s cancel:

$$Y_t - Y_s = e_{s+1} + e_{s+2} + \cdots + e_t \qquad (t - s \text{ terms}).$$

Step 3 – Second moment. Mean zero and uncorrelated terms, so

$$\mathbb{E}[(Y_t - Y_s)^2] = \mathrm{Var}(Y_t - Y_s) = (t - s)\sigma_e^2.$$

The same argument with roles swapped handles $t < s$; $t = s$ gives 0.

Answer

$\Gamma_{t,s} = \frac{1}{2}|t-s|\,\sigma_e^2$, which depends only on $|t-s|$. So the random walk is intrinsically stationary although it is not stationary – the semivariogram is a useful tool exactly for such processes.

Exercise 2.27

Exercise 2.27

For fixed $r \geq 1$ and constant ϕ: $Y_t = e_t + \phi e_{t-1} + \phi^2 e_{t-2} + \cdots + \phi^r e_{t-r} = \sum_{j=0}^{r} \phi^j e_{t-j}$. (a) Show stationarity for any ϕ. (b) Find the ACF.

(a)

Step 1 – Mean. $\mathbb{E}[Y_t] = \sum_{j=0}^{r} \phi^j \mathbb{E}[e_{t-j}] = 0$.

Step 2 – Covariance at lag $k \geq 0$.

$$\mathrm{Cov}(Y_t, Y_{t-k}) = \mathrm{Cov}\Big(\sum_{i=0}^{r} \phi^i e_{t-i},\ \sum_{j=0}^{r} \phi^j e_{t-k-j}\Big).$$

Indices match when $t - i = t - k - j$, i.e. $i = j + k$. Since $0 \leq i \leq r$ we need $0 \leq j \leq r - k$. Each match contributes $\phi^{j+k}\phi^j\sigma_e^2$:

$$\gamma_k = \sigma_e^2 \sum_{j=0}^{r-k} \phi^{j+k}\phi^j = \sigma_e^2\,\phi^k \sum_{j=0}^{r-k} \phi^{2j} \qquad (0 \leq k \leq r),$$

and $\gamma_k = 0$ for $k > r$ (no overlapping e's).

Step 3 – Why "any ϕ"? The sum is *finite* ($r+1$ terms), so it is always finite – no condition like $|\phi| < 1$ is needed (that would be required for an infinite sum, i.e. an AR(1)). And γ_k is free of t.

Answer

Mean 0 and γ_k depends only on k, so $\{Y_t\}$ is stationary for every ϕ (it is an MA(r) process).

(b)

Step 1 – Divide by γ_0.

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\phi^k \sum_{j=0}^{r-k} \phi^{2j}}{\sum_{j=0}^{r} \phi^{2j}}, \qquad 0 \leq k \leq r.$$

Step 2 – Closed form via geometric series. $\sum_{j=0}^{N} x^j = \frac{1-x^{N+1}}{1-x}$ with $x = \phi^2$:

$$\rho_k = \phi^k \, \frac{1-\phi^{2(r-k+1)}}{1-\phi^{2(r+1)}} \quad (\phi^2 \neq 1).$$

If $\phi = \pm 1$, every $\phi^{2j} = 1$, so the sums count terms: $\rho_k = \phi^k \dfrac{r-k+1}{r+1}$.

Answer

$$\rho_k = \begin{cases} \phi^k \, \dfrac{1-\phi^{2(r-k+1)}}{1-\phi^{2(r+1)}}, & 0 \leq k \leq r, \ \phi^2 \neq 1, \\ \phi^k \, \dfrac{r-k+1}{r+1}, & 0 \leq k \leq r, \ \phi = \pm 1, \\ 0, & k > r. \end{cases}$$

Sanity check: $\rho_0 = 1$. As $r \to \infty$ with $|\phi| < 1$, $\rho_k \to \phi^k$, the ACF of an AR(1): this process is a truncated AR(1).